Ericsson PE Node YEL-PE-11 Routing Protocol Flap SOP (ALARM_7365)

Node: YEL-PE-11 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_7365

Description

The Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on Ericsson PE Node YEL-PE-11 is a Critical failure domain that occurs due to the dynamic nature of routing protocols in the presence of network instability. The PE Node, acting as a border router, experiences a high volume of routing updates, causing the BGP routing table to churn, OSPF adjacency to be lost, and ISIS LSP flooding to occur. This phenomenon is exacerbated by the Ericsson equipment's reliance on complex routing protocol interactions, including route reflection, confederations, and multi-topology routing. The failure domain is further complicated by the node's role in the network, which necessitates a high degree of redundancy and failover capabilities. The ALARM_7365 code indicates a critical fault condition, requiring immediate attention to prevent network partitioning and service disruptions.

Pre-requisite Checks

- Verify that the Yelahanka Sub-station is accessible to remote hands personnel.
- Check the status of redundant links and ensure that they are functioning properly.
- Review the network topology and identify any potential bottlenecks or single points of failure.

Potential Root Causes

Cause	Probability
BGP route churn due to network instability	High
OSPF adjacency loss due to misconfiguration	Medium
ISIS LSP flooding due to hardware or software issues	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP summary information, Description: This command displays a summary of BGP peer connections, including the number of neighbors, the number of updates sent and received, and the number of prefixes advertised.

Command: show ospf neighbor, Expected Output: OSPF neighbor information, Description: This command displays information about OSPF neighbors, including their IP addresses, states, and timers.

Command: show isis database, Expected Output: ISIS LSP database information, Description: This command displays information about the ISIS LSP database, including the number of LSPs, the number of levels, and the number of segments.
```

Resolution Steps

Step 1: Step: Clear BGP session and re-establish, Command: clear bgp *, Description: This command clears the BGP session and re-establishes it, which may resolve issues related to BGP route churn.

Step 2: Step: Verify OSPF neighbor state, Command: show ospf neighbor, Description: This command verifies the state of OSPF neighbors and identifies any issues that may be contributing to adjacency loss.

Step 3: Step: Check ISIS LSP database for errors, Command: show isis database, Description: This command checks the ISIS LSP database for errors and identifies any issues that may be contributing to LSP flooding.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP summary information without errors, Description: This command verifies that the BGP session is stable and there are no errors.

Command: show ospf neighbor, Expected Output: OSPF neighbor information without errors, Description: This command verifies that the OSPF neighbor state is stable and there are no errors.

Command: show isis database, Expected Output: ISIS LSP database information without errors, Description: This command verifies that the ISIS LSP database is stable and there are no errors.
```

Roll-back Procedures

- Step: Restore previous BGP configuration, Command: load merge previous-bgp-config, Description: This command restores the previous BGP configuration, which may resolve issues related to BGP route churn.

- Step: Restore previous OSPF configuration, Command: load merge previous-ospf-config, Description: This command restores the previous OSPF configuration, which may resolve issues related to OSPF adjacency loss.

- Step: Restore previous ISIS configuration, Command: load merge previous-isis-config, Description: This command restores the previous ISIS configuration, which may resolve issues related to ISIS LSP flooding.